INCH-POUND

A-A-58076A
17 March 2022

SUPERSEDING
A-A-58076
02 August 1996

COMMERCIAL ITEM DESCRIPTION

CUTTER, COUNTERSINK

The General Services Administration has authorized the use of this commercial item description as a replacement for cutters class 1 and class 2 of MIL-H-4081B for all federal agencies.

1. SCOPE. This commercial item description (CID) covers countersinking cutters used on ferrous and non-ferrous materials.

2. CLASSIFICATION. The cutters shall conform to the following classes and styles.

2.1 Classes. The classes are as follows:

Class 1	Countersink, plain taper shank
Class 2	Countersink, external threaded shank

2.2 Styles. The styles are as follows:

Style A	With integral constructed pilot
Style B	With detachable drill for pilot

3. SALIENT CHARACTERISTICS.

3.1 Design. The cutters shall be made of heat treated high speed tool steel. The pilot diameter, taper shank or cutter centering angle, as applicable, and the cutting edges shall be concentric to 0.003 inch total indicator reading (TIR). Configuration and dimensional requirements shall be as shown (see Figure 1) and as specified in Table I for combination drill and countersink using Style A pilots. Countersink cutters using Style B pilots may deviate from pilots sizes shown in Table I, but shall have provisions for securing the pilots in place.

Beneficial comments, recommendations, additions deletions, clarifications, etc, and any data which may improve this document should be sent to: AFLCMC/EZSS, Bldg. 28, 2145 Monahan Way, Wright-Patterson AFB, OH 45433-7017 or email engineering.standards@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5133

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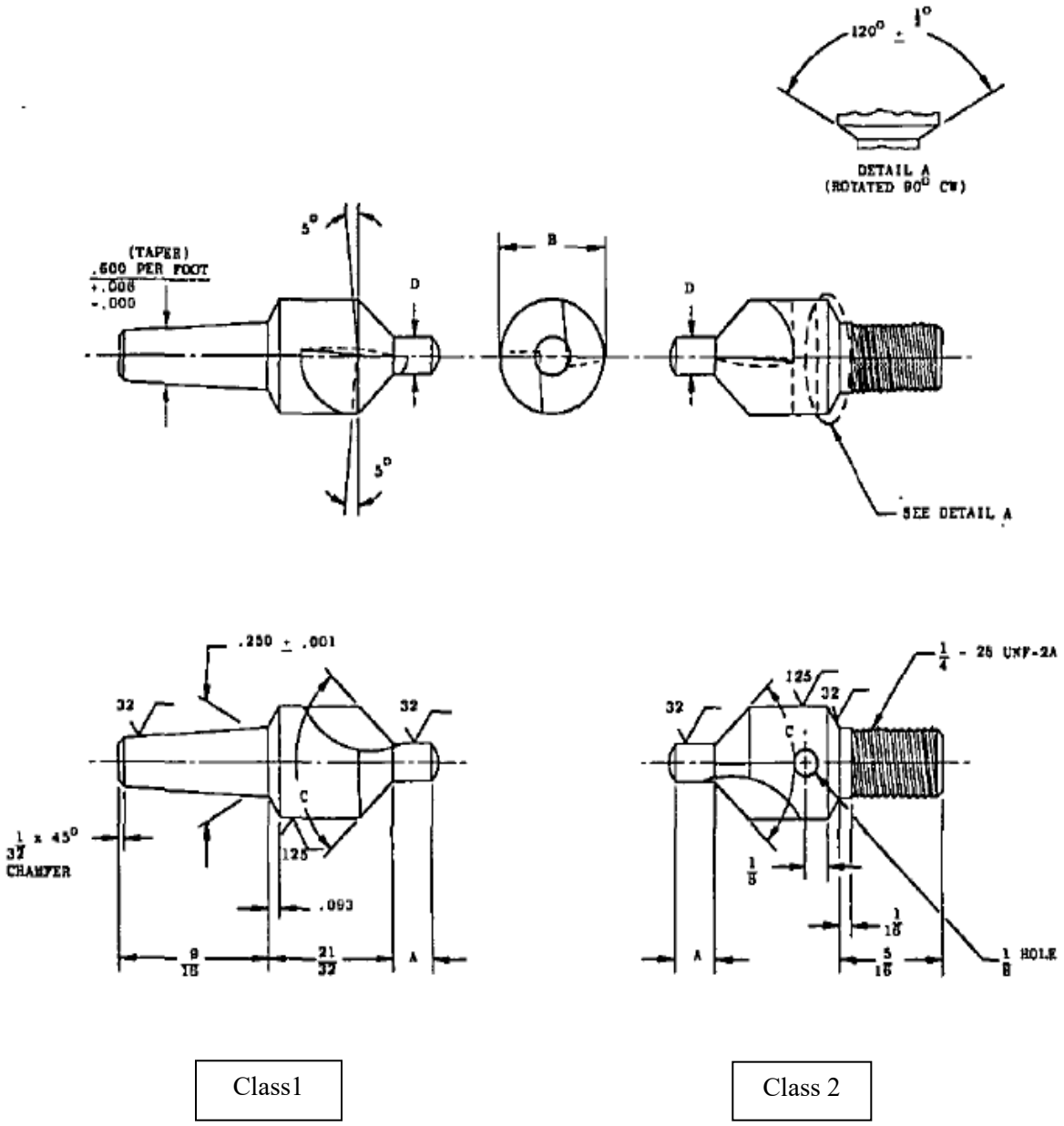
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TABLE I. Dimensional requirements.

Size	A - Pilot Exposed Length (Inches) ±1/64	B - Body Diameter (Inches) ±0.005	D - Pilot Diameter (Inches) ±0.000, ±0.001	C- Cutter angle (Degrees) ±0 degrees (°) 30’
A	3/32	0.437	0.093	60, 78, 82, 100, 110, 115
B	1/8	0.437	0.125	
C	5/32	0.437	0.156	
D	3/16	0.437	0.187	
E	1/8	0.500	0.125	
F	5/16	0.500	0.250	
G	5/16	0.625	0.312	
H	3/32	0.375	0.093	
I	1/8	0.375	0.125	
J	5/32	0.375	0.156	
K	3/16	0.375	0.187	
L	1/4	0.500	0.156	
M	1/4	0.875	0.250	
N	1/4	0.875	0.312	
O	5/16	0.875	0.375	

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FIGURE 1. Class 1 and 2 cutters.



Class1

Class 2

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3.2 Threads. Threads shall be 1/4-28 UNF-2A. All threads shall be in the inch system and shall conform to the class 2 tolerance limits specified in FED-STD-H28/2.

3.3 Identification marking. The countersink shall be marked with the manufacturers name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All markings shall be engraved, etched, molded, or indented directly on the items surface in such a manner that it remains clearly legible throughout the life of the item.

3.4 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, dry adherent, continuous, and not stained or discolored external surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections.

3.5 Cutting requirements. The cutter shall be capable of cutting repetitively when the cutter is mounted in a holder conforming to A-A-58075 using a portable, electric, or pneumatic powered hand tool. The holder and countersink shall be subjected to not less than ten repetitive countersink operation in stainless steel sheet, type 302, 8 gauge minimum thickness. The countersink shall be at least 0.406 ±0.005 in major diameter for all holes and the countersink surfaces shall not be rougher than 32 (AA). The included countersink angle and relation of an angle to the reference plane shall be accurate ±1/2 degree. Failure of the cutting requirement shall be cause for rejection of the representative lot.

4. REGULATORY REQUIREMENTS.

4.1 Recovered, recycled, or biobased materials. The offerer/contractor is encouraged to use recovered, recycled, and biobased materials to the maximum extent practicable, in accordance with Federal Acquisition Regulation (FAR), parts 23.403.

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they meet or exceed the requirements specified in this document.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1. Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producers own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2. Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

5.2.1 The company producing the item must have been producing a product meeting the requirements of this CID, for at least two years.

5.2.2 The company must have sold 1000 units meeting this CID, in the commercial market place over the past two years.

5.3 Performance tests.

5.3.1 Visual and dimensional inspection. Each sample selected shall be visually and dimensionally inspected for conformance with this CID.

5.3.2 Countersink cutting test. The countersink cutting test shall be performed with the cutter mounted in a holder conforming to A-A-58075, using a portable, electrical, or pneumatic powered hand tool. Holder and countersink shall be subjected to not less than ten repetitive countersink operations in stainless steel sheet, type 302, 8 gauge minimum thickness. The countersink shall be at least 0.406 ±0.005 inch major diameter for all holes and the countersink surfaces shall not be rougher than 32(AA). The included countersink angle and the relation of angle to the reference plane shall be accurate to ±1/2 degree. Failure of performance tests shall be cause for rejection of the representative lot. The number of cutters tested, and lot size will be specified in the contract.

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5.3.3 Additional requirements. Any additional requirements concerning responsibility for inspection, bid samples, or first article samples, shall be stated in the contract.

6. PACKAGING.

6.1 Preservation, packing, and marking. Requirements for preservation, packing, and marking of containers shall be as specified in the contract or order.

7. NOTES.

7.1. Source of documents.

7.1.1 FAR. Electronic copies of the FAR may be obtained from https://www.acquisition.gov.

7.1.2 Government documents. Specifications, standards, and handbooks referenced herein may be obtained at https://quicksearch.dla.mil/.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

a. Title, number, and date of this CID.
b. When applicable, the appropriate NSN.
c. When applicable, the specific classes, styles, and sizes (see Table I).
d. The number of bid samples required.
e. Applicable first article sampling requirements.
f. Applicable lot sampling requirements.
g. Required preservation, packing, and marking requirements.

7.3 National stock numbers (NSNs). The following NSNs are associated with this document. This list may not be indicative of all possible NSNs associated with this document.

SIZE	NSN 5133-00	CLASS	STYLE	CUTTER ANGLE
B	243-2298	1	A	100
C	243-2299	1	A	100
D	243-2300	1	A	100
M	293-2301	1	A	100
N	293-2303	1	A	100
O	595-8219	2	A	100
F	062-5290	2	A	100
F	062-5295	2	A	100
F	062-7216	2	A	100
A	177-7332	2	A	100
C	177-7334	2	A	100
D	177-7335	2	A	100

7.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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MILITARY INTERESTS:

Custodians:
Air Force - 184
Army – GL
DLA – GS

Reviewers:
Army – AV
GSA - FAS

CIVIL AGENCY COORDINATING ACTIVITY:

Preparing activity:
Air Force - 184

Agent:
Air Force - 110
(Project: 5133-2022-001)

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